$$x_2 - x_1$$

$$y_2 - y_1$$

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DISTANCE IN KNN (EUCLIDEAN DISTANCE)

For two points in a plane with coordinates (x_1, y_1) and (x_2, y_2), the Euclidean distance between them is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(x_1, y_1)

(x_2, y_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

, the distance is:

and

Example–For two points in 3D space

$$(x_1, y_1, z_1)$$

$$(x_2, y_2, z_2)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

More features in your data

$$x_2 - x_1$$

$$y_2 - y_1$$

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(x_1, y_1)

(x_2, y_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$